

You are holding steady right now

- ◆ Your temperature, blood sugar, and water are all being held near a **set point** — automatically.
- ◆ Anatomy asks *what* you're made of; physiology asks *how* it works.
- ◆ Today: the 6 levels you're built from, and the loop that keeps you stable.

Two words that split the whole course

- ◆ **Anatomy** = structure — what a part IS.
- ◆ **Physiology** = function — what a part DOES.
- ◆ Every unit this year is one of those two questions about a body system.

Today's words

Tissue — a group of similar cells with one job

Organ — two or more tissues working together

Organ system — organs that share a function

Homeostasis — a stable internal environment

Six levels, small to large

- ◆ Chemical → **Cell** → **Tissue** → **Organ** → **Organ system** → **Organism**.
- ◆ Each level is built from the one below it.
- ◆ A tissue = same cells; an organ = different tissues together.

Homeostasis = holding the set point

- ◆ The body keeps conditions (temp, water, glucose) near a **set point**.
- ◆ Drift off it, and a **negative feedback** loop pulls you back.
- ◆ Four parts: **stimulus** → **receptor** → **control center** → **effector**.

WATCH OUT

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Negative doesn't mean bad

COMMON MISTAKE

'Negative feedback' means the response goes in the OPPOSITE direction of the change — it REVERSES it. Sweating when hot is negative feedback, and it's good for you.

KEEP IN MIND

Negative = reverses. Positive = amplifies (rarer, e.g. childbirth).

YOU TRY

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Your turn

YOU TRY

- (a) Order: organ, cell, organism, tissue, organ system, chemical.
- (b) Is shivering negative or positive feedback?